METHODOLOGY RECAP · MODULE M3

Service utilization

What the module does · How to read its outputs

WHAT IT DOES

This module looks at whether health services are going **up, down, or holding steady** — and, crucially, whether a change is a **real disruption** or just normal monthly noise.

THE QUESTION IT ANSWERS

"Did something actually disrupt services here, or is this just the usual ups and downs?"

BUILT ON CLEAN DATA

It runs on the **adjusted** data from the previous module, so outliers and missing months don't create false alarms.

How it works

You can't judge a disruption by comparing one month to the last — services naturally rise and fall with the seasons and drift over the years. So for each facility (or area) and indicator, FASTR first builds an **expected** level: a line that already accounts for the long-term trend **and** the seasonal pattern.

Then it compares the **actual** reported value to that expected line. A month is flagged as a disruption when the actual strays too far from expected:

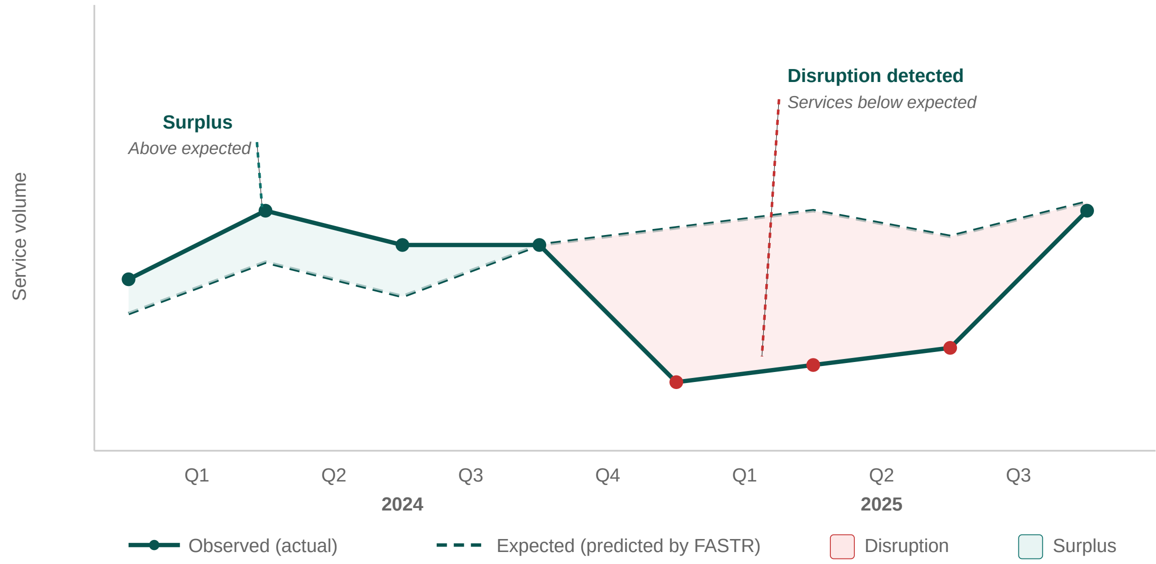
- a **sharp** drop or spike in a single month
- a **sustained** dip or surge that lasts several months
- a **run of missing** reports

Finally, a regression measures **how big** the disruption was — the average % below or above expected — and whether it's statistically real rather than chance. That's what lets you say *"ANC1 ran about 15% below expected from March to July."*

> ***The skill is the comparison: not "is this number high?" but "is it higher or lower than we'd expect for this place, this service, this time of year?"***

HOW IT WORKS · VISUALLY

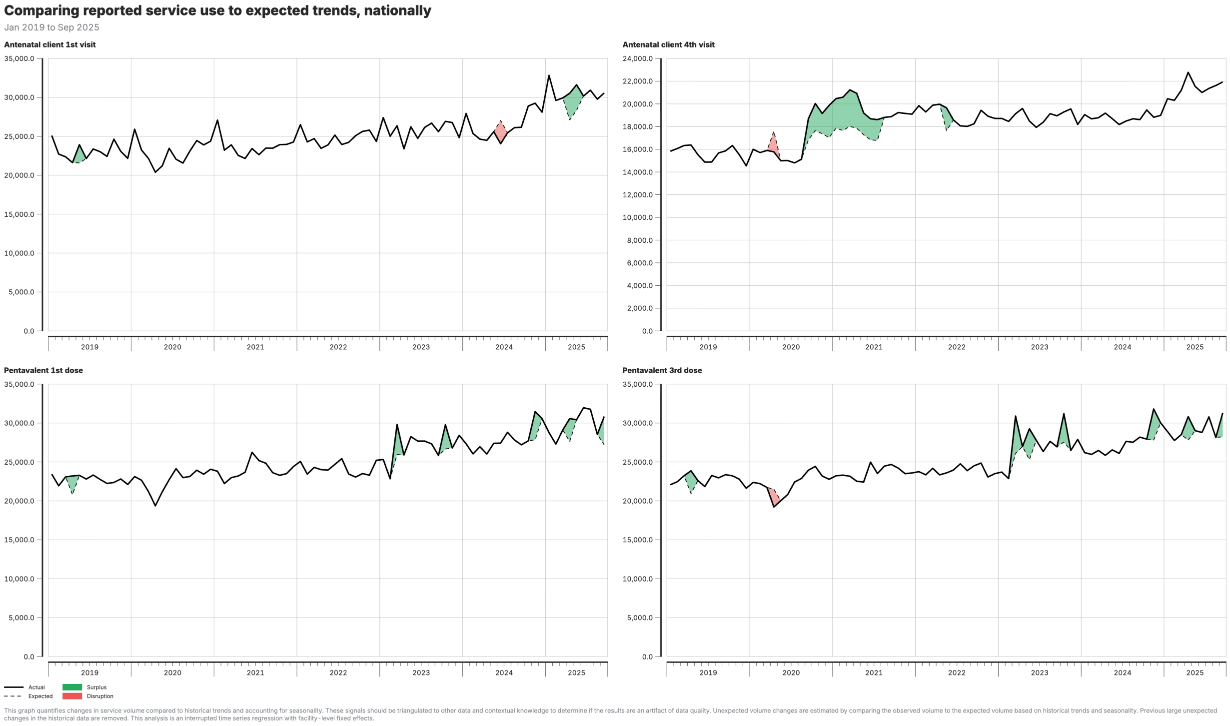
Spotting a disruption



The shaded gap is the distance between what actually happened and what FASTR expected — green above the line, red below.

OUTPUT 1 · THE TREND

Actual vs expected — spotting disruptions



- **What it shows** — one panel per indicator. The **black line** is the actual reported volume each month. Behind it sits the **expected** level FASTR computed for that area — a path that already builds in the long-term trend *and* the seasonal pattern, so "expected" means *normal for this place, this service, this time of year*. Where actual and expected pull apart, the gap is shaded: **green** when actual runs above expected (a surplus), **red** when it runs below (a disruption)
- **How to read it** — in three passes. **(1) Shape:** follow the black line for the overall trajectory — rising, flat, or falling across the years. **(2) Breaks:** scan for shaded patches; each is a stretch where reality left the expected path, and the **bigger and longer** the patch, the more serious — a deep red block over several months is a sustained disruption, a thin sliver is minor. **(3) Across indicators:** if red appears in the *same* months on several panels, something hit the whole system (a strike, a stock-out, a shock); red on one panel only points to a cause specific to that service

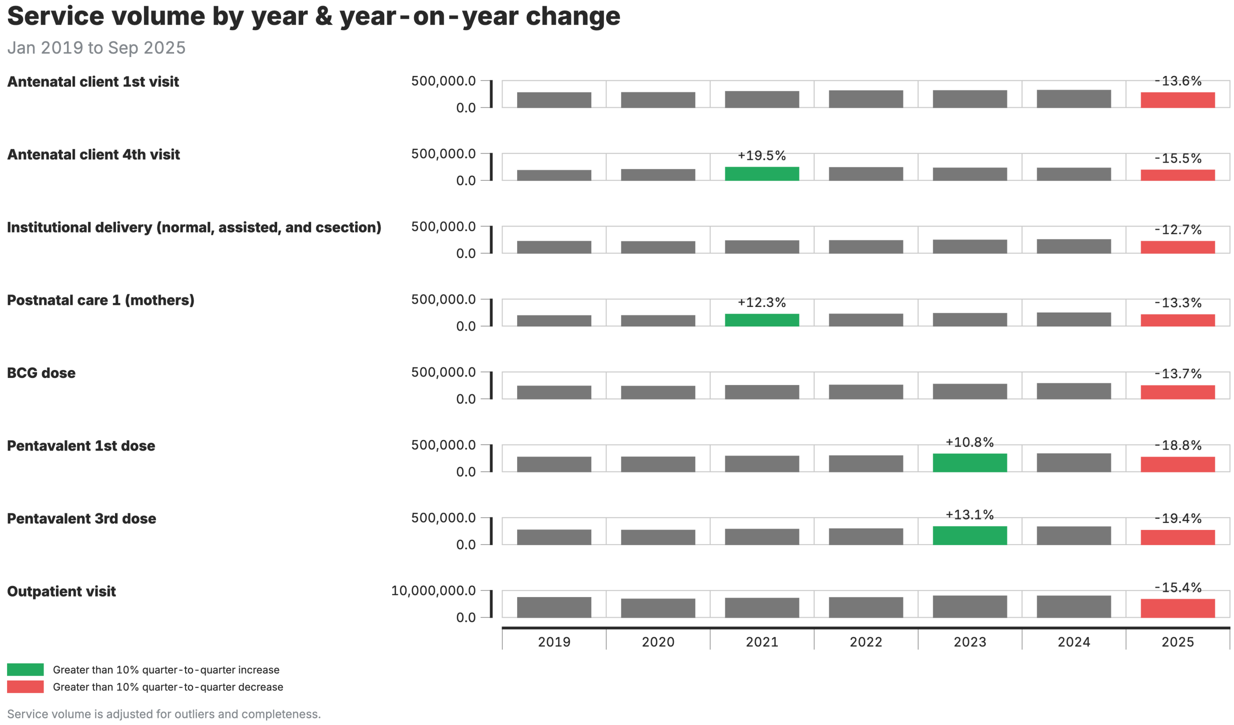
OUTPUT 1 · THE TREND (CONTINUED)

- **Watch for** — a single odd month is usually noise; wait for a sustained run before acting. Green isn't automatically good (it can be a catch-up campaign or double-counting) and red isn't automatically bad — both deserve a "why?". Line the red patches up against known events (funding cuts, elections, epidemic waves) to move from "something changed" to "this is what changed it"

> ***Worked example — in the* Antenatal 4th visit *panel, the green block across 2020–2021 is a long stretch where visits ran above the expected line. The small red marks near 2019 are short, one-off dips — not a sustained disruption.***

OUTPUT 2 · THE SIZE

Year-on-year change — how much moved



- **What it shows** — each indicator's **total annual volume** as bars across the years, with the year-on-year change labeled. A bar turns **green** when volume rose more than 10% on the year before, **red** when it fell more than 10%, and stays grey when it held roughly steady
- **How to read it** — this is the bird's-eye companion to the trend chart: that one shows *within-year* timing, this shows *between-year* magnitude. Read across a row to see whether a service is growing, shrinking, or stable, and read the labeled % on any colored bar to size the move. A **red bar in the most recent year** is the one to act on — the service ended the period materially lower than it started
- **Why you can trust it** — it's built on the adjusted data, so a red bar is a real fall in services, not an artefact of a missing month or a removed spike

> ***Worked example — Antenatal 1st visit ends 2025 at -13.6% (red): it finished the period well below the year before. An earlier green bar (+19.5%) was a rebound; the recent red is the one to act on. Together, the trend chart shows when and where services broke from expectation, and this one shows how much they moved.***